

الإجابات على "كراس الإجابة"

I- **Choose the correct answer : (1pt/correct answer)**

- 1- If a device fails to recognize the file format of incoming data, which OSI layer is most likely malfunctioning?
 - a- Application.
 - b- Presentation.
 - c- Session.
 - d- Transport.
- 2- What is a crossover cable used for?
 - a- To connect a PC to a wall outlet.
 - b- To connect two similar devices directly (e.g., PC to PC).
 - c- To connect two different devices directly (e.g., PC to Switch).
 - d- To connect to a printer.
- 3- Why is token passing used in ring topologies instead of contention-based access methods like CSMA/CD?
 - a- It is simpler to implement.
 - b- It prevents packet collisions entirely.
 - c- It allows higher transmission speeds.
 - d- It works only with wireless networks.
- 4- What makes logical and physical topologies different, particularly in Ethernet?
 - a- Logical defines bandwidth, physical defines cable length.
 - b- Logical describes the actual wiring layout.
 - c- Logical defines data flow; physical defines layout.
 - d- They are always the same in modern networks.
- 5- What is the subnet mask for a network with 14 usable hosts?
 - a- 255.255.255.240
 - b- 255.255.255.248
 - c- 255.255.255.252
 - d- 255.255.255.255

II- **Treat 3 of the following 4 questions: (5 pts/question)**

- 1- Explain the concepts of error detection and error correction in computer networks. Which OSI layers are primarily responsible for these functions, and what techniques or mechanisms do they use to ensure data integrity during transmission?
- 2-
 - a- Define physical topology in networking. How does it differ from logical topology? Provide examples of each.
 - b- Compare the physical star, bus, and ring topologies in terms of cabling cost, ease of troubleshooting, and fault tolerance?
 - c- In a ring topology, what is the importance of token passing? How does it affect network communication and collision management?
- 3- You are a network administrator for a small company that has been assigned the IP block 192.168.10.0/24
 - a- What is the new subnet mask in dotted-decimal notation required to divide the network into four equal subnets?
 - b- How many valid hosts can each subnet support?
 - c- List the network address, first host, last host, and broadcast address for each subnet.
 - d- If a device has the IP address 192.168.10.129, which subnet does it belong to? Justify your answer.
- 4- You have been allocated the IP block 210.0.0.0/24 to set up a network for an organization with three departments:
HR: 60 devices
IT: 50 devices
Admin: 40 devices
 - a- What is the minimum subnet size (in number of hosts) that will satisfy all the department's needs?
 - b- What subnet mask would you choose to meet this requirement?
 - c- How many total subnets will this mask create?
 - d- How many IP addresses are wasted across all assigned subnets?
 - e- Can you add a fourth department needing 60 hosts with this subnet mask? Explain your answer.

الإمضاء	الإسم	إسم اللجنة
	فاروق الحركة	رئيس اللجنة
	أنطون يزبك	نائب الرئيس
		لجنة المعلوماتية

التاريخ : / / 2025/

أسس التصحيح للمسابقة : الشبكات. المعلوماتية الإدارية (23102)

I.

1	2	3	4	5
b	b	b	a	a

(5pts)

II.

1.

Error Detection means *finding out* if the transmitted data has been altered or corrupted during transmission. It doesn't fix the problem itself, it just raises the red flag: "something went wrong."

Error Correction goes a step further. It not only detects the error but also *restores* the original data, either by asking for a retransmission or by fixing the error using redundant information.

OSI Layers Responsible

1. Data Link Layer (Layer 2)

- Ensures error detection (and sometimes correction) for frames transmitted between directly connected nodes.
- Uses mechanisms like:
 - Parity bits (very basic check)
 - Checksums
 - Cyclic Redundancy Check (CRC) → the most common at this layer.

2. Transport Layer (Layer 4)

- Focuses on end-to-end reliability between two applications.
- Uses:
 - Checksums (TCP, UDP headers include a checksum to detect errors in the segment).

How Data Integrity is ensured

- At the **Data Link Layer**, errors are detected quickly at each hop using CRC. If corrupted, the frame is discarded and retransmitted by lower-level protocols.
- At the **Transport Layer**, end-to-end checksums plus retransmission strategies guarantee that the final data delivered to the application is correct and complete.

2.

a.

Physical Topology

Refers to the *actual physical layout* of devices, cables, and connections in a network.

Example: In a **star topology**, all computers connect to a central switch or hub with separate cables.

الإمضاء	أسماء المشاركين في التصحيح	الإمضاء	أسماء المشاركين في التصحيح

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	فارق الحركة	رئيس اللجنة
	أنطون يزبك	نائب الرئيس
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• **Logical Topology**

Refers to the way data flows within the network, regardless of its physical layout. It's more about the path of communication than the cable arrangement.

Example: Token Ring networks (IBM). The devices were physically wired in a star to a hub called a MAU (Multistation Access Unit), but logically, the data moved in a circular path from one device to the next.

• **Key difference:**

Physical = how it looks and is wired.

Logical = how data moves and is controlled.

b.

Topology	Cabling Cost	Ease of Troubleshooting	Fault Tolerance
Star	Higher (each device needs its own cable to the central hub/switch)	Easy (a problem usually affects only one link)	High (failure of one cable/device doesn't affect the others, but failure of the central hub/switch brings the network down)
Bus	Low (all devices share one backbone cable)	Harder (a fault anywhere along the backbone can take down the whole network)	Low (single cable failure affects all devices)
Ring	Medium (each device connects to two others, forming a circle)	Moderate (fault isolation is tricky, though some rings have dual-ring redundancy)	Medium to low (a break can disrupt the whole network unless fault-tolerant rings are used)

c.

- In a **ring topology**, data travels in one direction from device to device.
- To prevent multiple devices from transmitting at the same time (which would cause collisions), a **token**—a special control frame—circulates around the ring.

Importance:

- **Collision Management:** Only the device holding the token can transmit, so collisions are eliminated.
- **Fair Access:** Each device gets a turn when the token arrives; no single device can monopolize the medium.
- **Efficient Communication:** Since no collisions occur, the bandwidth is used more predictably.

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الإمضاء	الإسم	إسم اللجنة
	فاروق الحركة	رئيس اللجنة
	أنطون يزبك	نائب الرئيس
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3.

- a. 255.255.255.192
b. The valid hosts in each subnet: $2^6 - 2 = 62$ hosts
c.

Network address	First host	Last host	Broadcast
192.168.10.0	192.168.10.1	192.168.10.62	192.168.10.63
192.168.10.64	192.168.10.65	192.168.10.126	192.168.10.127
192.168.10.128	192.168.10.129	192.168.10.190	192.168.10.191
192.168.10.192	192.168.10.193	192.168.10.254	192.168.10.255

- d. Belong to the subnet 192.168.10.128
192.168.10.129 AND with the mask
255.255.255.192
192.168.10.128

4.

- a. Min in number of host: 62 hosts
b. 255.255.255.192
c. 4 subnets
d. Across all assigned subnet the wasted IP addresses =
 $2 + 12 + 22 + 62 = 98$ IP
e. Yes each subnet supports 62 hosts
with the mask 255.255.255.192 we have 2 bits to the network and 6 bits to the hosts
it means we have $2^6 - 2 = 62$ hosts in each subnet.

1 pt each = 5 pts

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	سليم	سليم	سليم
	الهادي	الهادي	الهادي
	سليم	سليم	سليم